Precious Fikile Mkhize 0210210785080

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INVESTIGATE THE USE PF COMPUTER TECHNOLOGY IN AN ORGANISATION

US 14963

SECTION A: FORMATIVE ASSESSMENT

Question 1 (SO 1, AC 1, AC 2, AC 3, AC 5; Range) (10)

An investigation plan is a roadmap for the investigation. It should cover the following:

1. Purpose / Objectives: Why the investigation is being done. For example to find out what computer technology is used, how effective it is and what gaps exist.

2. Scope: Which departments, systems and time period will be covered.

3. Background of the Organisation: Name, size, type of business and location.

4. Methodology: How data will be collected - interviews, questionnaires, observation and document analysis.

5. Target Population: Who will be interviewed - managers, IT staff and end-users.

6. Resources Needed: People, computers, internet, transport, stationery and budget.

7. Time Frame / Schedule: Start and end dates for each activity.

8. Data Collection Tools: Interview schedules, checklists and survey forms.

9. Data Analysis Method: How the data will be analysed - qualitatively and quantitatively.

10. Ethical Considerations: Confidentiality, consent and permission from management.

11. Expected Outcomes / Deliverables: Final report and recommendations.

Question 2 (SO 1, AC 4) (5)

After drafting the investigation plan you should:

1. Review and validate the plan: Check for errors, gaps and whether it meets the objectives.

2. Get approval: Submit the plan to your supervisor, lecturer or organisation management for approval.

3. Pilot test: Test your questionnaires or interview questions on a small group to see if they are clear and understandable.

4. Make amendments: Correct and improve the plan based on feedback from the supervisor and pilot test.

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5. Prepare for implementation: Organise resources, set appointments with respondents, print data collection tools and get permission letters.

Question 3 (SO 2, AC 1, AC 3; Range) (10)

1. Define the problem: For example, the organisation has slow systems and you need to find out why.

2. Plan the investigation: Draft the investigation plan as described in Question 1.

3. Get permission: Get a letter from your college and permission from the organisation, for example Shoprite.

4. Collect data:

Interview - Ask the IT manager "What software do you use for stock control?"

Questionnaire - Give cashiers a form about POS system problems.

Observation - Watch how staff use the Pastel accounting system.

Document review - Check inventory reports and IT policy documents.

5. Analyse data: Sort and interpret responses, for example 70% of users say the internet is slow.

6. Compile findings: Write down what hardware, software and networks are used and how they are used.

7. Draw conclusions and recommendations: For example, recommend upgrading RAM and providing training.

8. Present and compile final report.

Question 4 (SO 3, AC 1; Range) (8)

1. Needs Assessment: Identify what the organisation needs, for example we need a faster payroll system.

2. Feasibility Study: Check if the project is affordable, technically possible and will benefit the organisation. This includes economic, technical and operational feasibility.

3. Setting Objectives: Set clear goals, for example "Reduce payroll processing time by 50%".

4. Budgeting: Allocate money for hardware, software, training and maintenance.

5. Resource Planning: Identify staff, equipment and vendors needed.

6. Risk Assessment: Identify risks like data loss, hacking and resistance from staff.

7. Timeline / Project Scheduling: Use Gantt charts to set milestones and deadlines.

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8. Procurement Plan: Process of selecting suppliers and buying equipment.

9. Training Plan: Plan how users will be trained to use the new system.

10. Implementation Strategy: Decide on method - direct changeover, parallel, phased or pilot implementation.

Question 5 (SO 3, AC 2) (8)

1. Transaction Processing System (TPS): Handles daily routine transactions. Example is Point of Sale (POS) system at Pick n Pay, payroll system and Pastel for invoicing.

2. Management Information System (MIS): Produces reports for middle management using data from TPS. Example is monthly sales reports, stock level reports and budget reports.

3. Decision Support System (DSS): Helps top management make non-routine decisions by analysing data. Example is analysing sales data to decide whether to open a new branch.

4. Office Automation System (OAS): Supports office work such as MS Office, email and Teams.

5. Enterprise Resource Planning (ERP): Integrated system like SAP that manages all departments in one system.

Question 6 (SO 3, AC 3) (3)

1. Systems Analyst - Analyses user needs and designs the system.

2. Programmer / Software Developer - Writes and customises software.

3. Network Administrator - Sets up and maintains the network and internet.

4. IT Manager / CIO - Oversees IT projects and strategy.

5. Database Administrator - Manages organisation's databases.

6. End-Users - Staff who use the system such as clerks and HR staff.

7. Technician / Helpdesk Support - Installs hardware and provides technical support.

Question 7 (SO 4, AC 1, AC 2) (12)

Strengths:

1. Speed and efficiency - Processes payroll, invoices and reports faster than manual.

2. Accuracy - Reduces human errors in calculations and data capturing.

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3. Storage - Stores large amounts of data safely.

4. Communication - Email, WhatsApp and Teams improves internal and external communication.

5. Better customer service - Quick access to customer information.

6. Improved decision making - Managers get accurate reports on time.

Weaknesses:

1. High cost - Purchase of hardware, software licences and maintenance is expensive.

2. Security risks - Risk of viruses, hacking and data theft.

3. Dependence on technology - When system is down, work stops.

4. Need for training - Some staff lack computer skills and need constant training.

5. Job losses - Some jobs are automated and replaced by computers.

6. Outdated hardware/software - Slow computers and old Windows versions affect performance.

7. Power failure - Load shedding and power cuts affect computer systems.

Question 8 (SO 5, AC 1, AC 2, AC 3) (6)

1. Audience: Consider who you are presenting to and use simple language if they are non-IT managers.

2. Format of presentation: Use clear headings, tables, charts and PowerPoint.

3. Accuracy: Present facts only, supported by data collected.

4. Confidentiality: Do not reveal sensitive company information.

5. Clarity and conciseness: Be brief, straight to the point and avoid technical jargon.

6. Visual aids: Use graphs showing usage of systems and problems found.

7. Recommendations: Must be practical, realistic and linked to findings.

8. Time management: Stick to allocated time and allow time for questions.

9. Professionalism: Professional dress, language and prepare handouts.